A monic resultant for Bessel derivative coefficients

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Abstract

For the differential equation $f'' + cx^{-1}f' + (1 + d(c-d-1)x^{-2})f = 0$, we express each derivative $f^{(n)}$ in the basis (f, f') using two polynomial coefficients. Their resultant with respect to x is a monic polynomial in c over $\mathbb{Z}[d]$, of degree $n(n-1)$, for every $n \geq 2$. The specialization $d = 0$, $c = 2\nu + 1$ proves the degree assertion formulated in Section 2.3(a) of Lorch and Muldoon (1995) for derivatives of $x^{-\nu}J_\nu(x)$. A classical theorem of Siegel then implies that all nonzero zeros of these derivatives are transcendental whenever $\nu \in \mathbb{Q} \setminus \frac{1}{2}\mathbb{Z}$. More generally, the same conclusion holds for $x^{-\mu}J_\nu(x)$ when $\mu - \nu$ or $\mu + \nu$ is an algebraic integer. The remaining rational-root classification and the all-orders question for derivatives of the unnormalized function J_ν are not resolved here.

1 The coefficient resultant

Lorch and Muldoon [1, Section 2.3] studied the algebraic zeros of successive derivatives of the normalized Bessel function $x^{-\nu}J_\nu(x)$. Their polynomial elimination procedure led to three conjectures. The first concerns only the degree of a resultant: they proposed that its degree in ν is $n(n-1)$ for derivative order n. We prove this assertion by identifying the highest-weight coefficient polynomials, and obtain the stronger fact that a suitable affine change of parameter makes the resultant monic.

The proof is algebraic. The only transcendence input, used subsequently for zeros, is the classical nonvanishing and logarithmic-derivative theorem for Bessel functions of rational order. In particular, no new algebraic-independence theorem is required for the resultant calculation.

Let c, d be indeterminates, and write

$$\Delta = d(c-d-1).$$

Consider formally the second-order equation

$$f'' + \frac{c}{x}f' + \left(1 + \frac{\Delta}{x^2}\right) f = 0. \tag{1}$$

Define $A_n, B_n \in \mathbb{Z}[d, c, x]$ for $n \geq 1$ by

$$\begin{aligned} A_1 &= 0, \qquad B_1 = 1, \\ A_{n+1} &= x\partial_x A_n - nA_n - (x^2 + \Delta)B_n, \\ B_{n+1} &= A_n + x\partial_x B_n + (1 - c - n)B_n. \end{aligned} \tag{2}$$

Differentiating and substituting (1) gives

$$f^{(n)} = x^{-n}(A_n f + xB_n f') \tag{3}$$

for every local solution on a domain avoiding zero. Both coefficient polynomials are even in x.

Theorem 1.1. *For every integer $n \geq 2$, the polynomial*

$$R_n(c,d) = \mathrm{Res}_x\big(A_n(x,c,d), B_n(x,c,d)\big)$$

belongs to $\mathbb{Z}[d][c]$, is monic as a polynomial in c, and has c-degree $n(n-1)$.

Our convention is the usual resultant with respect to x, without removing factors corresponding to $x = 0$. Removing such factors could change the degree under discussion. At $d = 0$, the polynomials denoted by p_n, q_n in [1] satisfy

$$p_n = -B_n, \qquad q_n = -A_n.$$

Their x-degrees are even, so sign changes and interchanging the two arguments leave the resultant unchanged. Consequently Theorem 1.1 proves their stated degree conjecture in exactly their resultant convention. Since $c = 2\nu + 1$, the leading coefficient in their original parameter ν is $2^{n(n-1)}$.

2 The highest-weight calculation

Treat d as a coefficient, and assign weight one to both x and c. The additional potential Δ has weight at most one. The recurrences therefore give

$$\deg A_n \leq n, \qquad \deg B_n \leq n-1,$$

where these are total degrees in x, c. For $n \geq 2$, the exact x-degrees are

$$d_A = 2\lfloor n/2 \rfloor, \qquad d_B = 2\lfloor (n-1)/2 \rfloor. \tag{4}$$

We include the verification, as preservation of these degrees is needed when reducing a resultant modulo a prime.

Let a_n, b_n be the homogeneous parts of weights $n, n-1$ of A_n, B_n, respectively. Terms involving Δ have lower weight, and hence

$$a_n = -x^2 b_{n-1}, \qquad b_n = -c b_{n-1} - x^2 b_{n-2} \quad (n \geq 3), \qquad b_1 = 1, \quad b_2 = -c. \tag{5}$$

In particular, these highest-weight parts are independent of d. The recurrence, or induction using Pascal's identity, gives

$$b_n(x,c) = \sum_{j=0}^{\lfloor (n-1)/2 \rfloor} (-1)^{n-1-j} \binom{n-1-j}{j} c^{n-1-2j} x^{2j}. \tag{6}$$

The leading terms in this formula and $a_n = -x^2 b_{n-1}$ give the degrees in (4). The full recurrences imply the matching upper bounds on x-degree, so there are no further terms of higher x-degree in A_n or B_n.

Lemma 2.1. *For every $n \geq 2$,*

$$\mathrm{Res}_x\big(a_n(x,1), b_n(x,1)\big) = 1.$$

Proof. For $n = 2$, the pair is $(-x^2, -1)$ and the assertion is immediate. Suppose $n \geq 3$. We first check the absence of common finite zeros over an arbitrary field, including fields of positive characteristic. Since $b_n(0,1) = (-1)^{n-1}$, a common zero ξ cannot be zero. The equation $a_n(\xi, 1) = 0$

then gives $b_{n-1}(\xi,1)=0$. The recurrence (5), using $\xi\neq 0$, next gives $b_{n-2}(\xi,1)=0$. Continuing backwards eventually contradicts $b_1=1$.

It remains to handle the possible loss of degree after reduction modulo a prime. If n is even, the coefficient of the highest power of x in $a_n(x,1)$ is ± 1. If n is odd, the corresponding coefficient in $b_n(x,1)$ is ± 1. These statements follow directly from (6). Thus at least one of the two prescribed degrees d_A, d_B is preserved modulo every prime. Homogenize at precisely these degrees. The homogenizations have no common zero at infinity, and the preceding argument rules out a common finite zero. Their resultant is consequently nonzero modulo every prime.

The integer $\mathrm{Res}_x(a_n(x,1), b_n(x,1))$ therefore has no prime divisor, so it is 1 or -1. For polynomials in x^2, the identity

$$\mathrm{Res}_x(P(x^2),Q(x^2)) = \mathrm{Res}_z(P(z),Q(z))^2$$

shows that this integer is nonnegative. It must equal 1. □

Proof of Theorem 1.1. Introduce a new indeterminate T, substitute $x=TX, c=T$, and divide the two polynomials by their respective highest weights. In the polynomial ring in X with coefficients in $\mathbb{Z}[d][T^{-1}]$, we have

$$T^{-n}A_n(TX,T,d) = a_n(X,1)+O(T^{-1}),$$
$$T^{-(n-1)}B_n(TX,T,d) = b_n(X,1)+O(T^{-1}).$$

Here $O(T^{-1})$ means a polynomial in T^{-1} with zero constant term. These expressions have prescribed X-degrees d_A, d_B. The scaling laws for the resultant, followed by Lemma 2.1, give

$$R_n(T,d) = T^{nd_B+(n-1)d_A-d_Ad_B}\left(1+O(T^{-1})\right).$$

For $n=2m$ the exponent is $2m(2m-1)$, while for $n=2m+1$ it is $2m(2m+1)$. Thus it equals $n(n-1)$ in both cases. Since the resultant is in $\mathbb{Z}[d][T]$, its leading coefficient is exactly 1. □

3 Arithmetic consequences for zeros

For rational ν, the classical theorem of Siegel implies that $J_\nu(\alpha)\neq 0$ and $J'_\nu(\alpha)/J_\nu(\alpha)$ is transcendental whenever α is nonzero and algebraic; see [2, Chapter 6, Section 2], or the formulation used in [1]. For algebraic μ, a local branch of $f=x^{-\mu}J_\nu$ therefore satisfies

$$f(\alpha)\neq 0, \qquad \frac{f'(\alpha)}{f(\alpha)} = \frac{J'_\nu(\alpha)}{J_\nu(\alpha)} - \frac{\mu}{\alpha} \notin \overline{\mathbb{Q}}. \tag{7}$$

This input is independent of the branch used near α.

Put $d=\mu-\nu$ and $c=2\mu+1$. The transformed Bessel equation is exactly (1), since

$$\mu^2-\nu^2 = d(c-d-1).$$

If $f^{(n)}(\alpha)=0$ at a nonzero algebraic point, then (3) and (7) imply

$$A_n(\alpha,c,d) = B_n(\alpha,c,d) = 0. \tag{8}$$

Indeed, a nonzero coefficient B_n would express f'/f as an algebraic number; after $B_n=0$, the nonvanishing of f forces $A_n=0$. In particular, (8) gives $R_n(c,d)=0$.

Corollary 3.1. *Let $\nu \in \mathbb{Q} \setminus \frac{1}{2}\mathbb{Z}$. For every integer $n \geq 0$, every nonzero zero of*

$$\left(\frac{d}{dx}\right)^n \left(x^{-\nu} J_\nu(x)\right)$$

is transcendental.

Proof. The cases $n = 0, 1$ follow directly from (7). For $n \geq 2$, an algebraic exceptional point would make the rational number $c = 2\nu + 1$ a root of the monic integer polynomial $R_n(c, 0)$. The rational-root theorem would then imply $c \in \mathbb{Z}$, contrary to $2\nu \notin \mathbb{Z}$. □

Corollary 3.2 (Integral shifts)**.** *Let $\nu \in \mathbb{Q} \setminus \frac{1}{2}\mathbb{Z}$ and $\mu \in \overline{\mathbb{Q}}$. Suppose that at least one of $\mu - \nu$ and $\mu + \nu$ is an algebraic integer. For every $n \geq 0$, every nonzero zero of $(d/dx)^n(x^{-\mu} J_\nu(x))$ is transcendental.*

Proof. Suppose first that $d = \mu - \nu$ is an algebraic integer. For $n \geq 2$, an exceptional point would give $R_n(c, d) = 0$. Monicity makes c integral over $\mathbb{Z}[d]$, hence integral over $\mathbb{Z}$. The number $2\nu = c - 2d - 1$ would be a rational algebraic integer and therefore an integer, a contradiction. If $\mu + \nu$ is integral instead, the same argument uses the formal order $-\nu$ in the differential equation; the chosen solution and the input (7) are unchanged. The cases $n = 0, 1$ again follow directly from that input. □

Corollary 3.3 (Denominator obstruction)**.** *Let $\mu, \nu \in \mathbb{Q}$. If a derivative of $x^{-\mu} J_\nu(x)$ has a nonzero algebraic zero, every prime in the denominator of 2ν occurs in the denominator of $\mu - \nu$ and also in the denominator of $\mu + \nu$.*

Proof. The cases of derivative order zero or one have no exceptional point by (7), so suppose $n \geq 2$. If $d = \mu - \nu$ is integral at a prime p, then the coefficients of the monic polynomial $R_n(c, d)$ lie in $\mathbb{Z}_{(p)}$. The rational root c is integral over this integrally closed ring, so c and $2\nu = c - 2d - 1$ are p-integral. Replacing the formal order by $-\nu$ gives the other assertion. □

4 Examples and the remaining questions

The first two resultants provide short checks on the normalization. With $\Delta = d(c - d - 1)$, the recurrence gives

$$A_2 = -x^2 - \Delta, \qquad B_2 = -c, \qquad R_2(c, d) = c^2.$$

At the next order,

$$A_3 = cx^2 + (c + 2)\Delta, \qquad B_3 = c(c + 1) - \Delta - x^2,$$

and hence

$$R_3(c, d) = \left(c^2(c + 1) + 2\Delta\right)^2.$$

Their degrees in c are 2 and 6, respectively, as required. For $d = 0$, these become c^2 and $c^4(c + 1)^2$.

The passage from common roots to a resultant has an essential one-way qualification in the arithmetic application. Vanishing of $R_n(c, d)$ need not give a *nonzero* common root. For example, at $d = 0, c = 0$, the pair $(A_2, B_2) = (-x^2, 0)$ has only the common root zero. Every corollary above uses only the necessary implication

$$\text{nonzero algebraic exceptional point} \implies R_n(c, d) = 0.$$

At $d = 0$, one also has $A_n(0, c, 0) = 0$ and

$$B_n(0, c, 0) = (-1)^{n-1} c(c+1) \cdots (c+n-2).$$

Thus $c = 0, -1, \ldots, -(n-2)$ are genuine roots of the raw resultant, often recording an origin factor. Theorem 1.1 does not determine all remaining integer roots, or the greatest common divisor in x at those roots.

Specifically, the argument establishes the degree conjecture in [1, Section 2.3(a)]. The rational-root classifications proposed in their parts (b),(c) remain outside the proof. The condition $2\nu \in \mathbb{Z}$ includes integer orders as well as strictly half-integral orders, and leaves infinitely many possibilities to analyze.

The closing question of [1] for the unnormalized derivatives $J_\nu^{(n)}$ is a different problem. Their Theorem 6 covers $0 \le n \le 18$, with the nonzero algebraic exceptions $J_{\pm 1}^{(3)}(\pm\sqrt{3}) = 0$ and $J_0^{(4)}(\pm\sqrt{3}) = 0$. In the present general parametrization, the unnormalized function has $\mu = 0$ and $d = -\nu$; the denominator obstruction then gives no new restriction. No assertion about the absence of further unnormalized exceptions is made here.

The explicit match with the 1995 conjecture is a statement about what the proof establishes. A claim of priority over all intervening literature would require a separate, exhaustive bibliographic review.

References

[1] L. Lorch and M. E. Muldoon, *Transcendentality of zeros of higher derivatives of functions involving Bessel functions*, International Journal of Mathematics and Mathematical Sciences **18** (1995), no. 3, 551–560. doi:10.1155/S0161171295000706.

[2] A. B. Shidlovskii, *Transcendental Numbers*, translated by N. Koblitz, de Gruyter Studies in Mathematics, vol. 12, Walter de Gruyter, Berlin, 1989.